

# **Estimation and hypothesis testing**

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# Objectives

At the end of this course students will be able to:

- Define Inferential statistics
- Know statistical estimation
- Understand hypothesis testing & the “types of errors” in decision making.
- Use test statistics to examine hypothesis about population parameter

# Inference



Use a random sample to learn something about a larger population

# Inferential Statistics

- **Inferential Statistics:** Are statistical methods used for drawing conclusions about a **population** based on the information obtained from the a **sample** of observations drawn from that population

# Inferential Statistics

- Involves

- Estimation
- Hypothesis testing

- Purpose

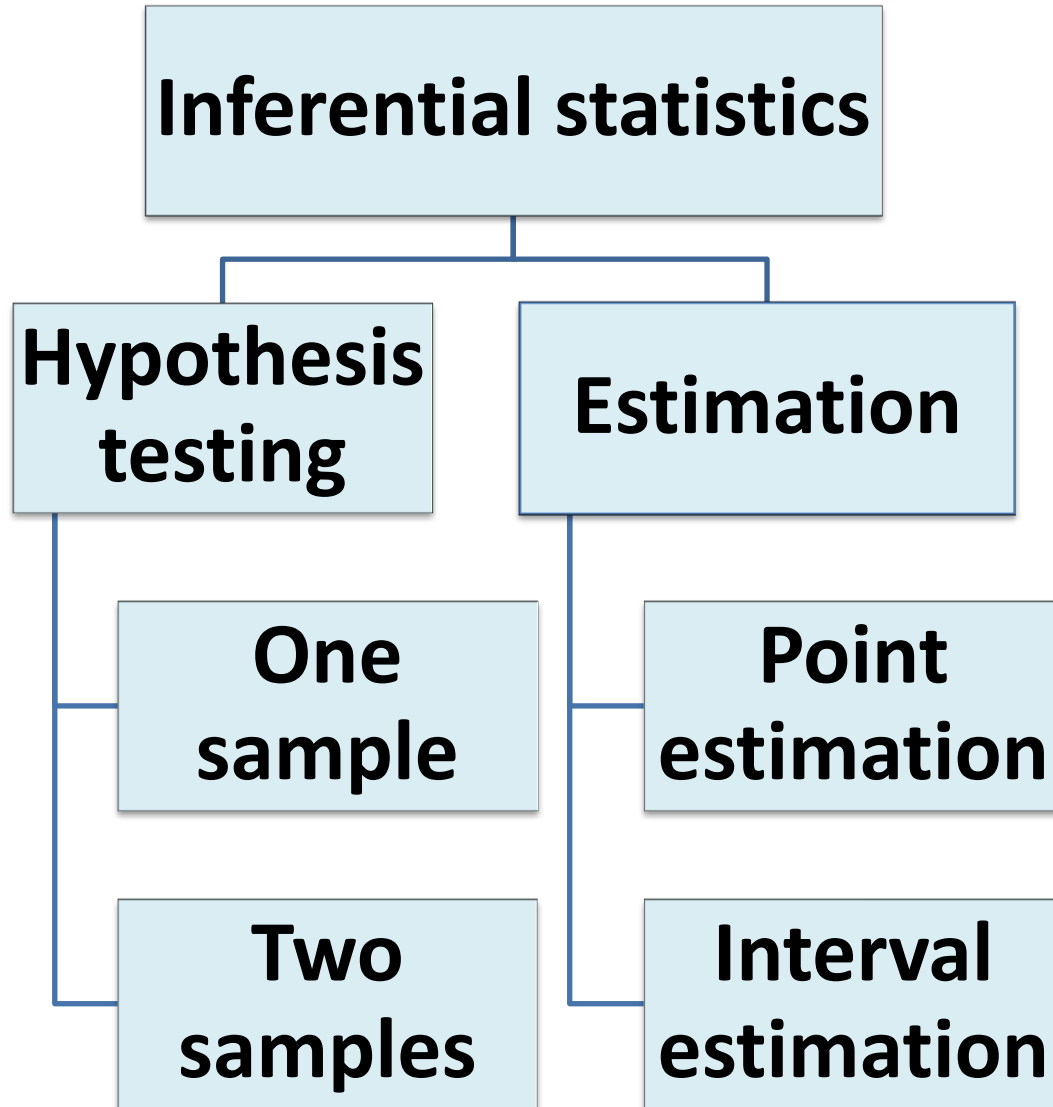
- Make decisions about population characteristics



The diagram consists of two overlapping rectangular boxes. The top box is white with a thin black border and contains the text "This image cannot currently be displayed." in a small font. The bottom box is also white with a thin black border and contains the same text. A black speech bubble with a white border points from the top box towards the bottom box. Inside the speech bubble, the word "Population?" is written in a large, bold, white sans-serif font.

**Population?**

# Inferential Statistics



# Inferential process

**Estimate  
& test  
population  
parameter**



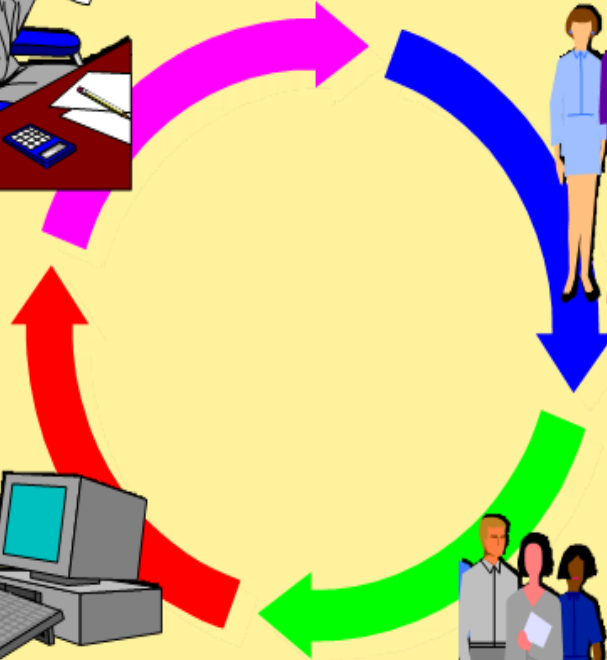
**Population**



**Sample  
statistic**



**Sample**



# Statistical Estimation

- Estimation is the process of determining a likely value of population parameter, based on information collected from the sample.
- Estimation is the use of sample statistics to estimate the corresponding population parameters
- The objective of estimation is to determine the ***approximate value*** of unknown population parameter on the basis of a sample statistic



- We assumed that the number of neutrophils in a sample of 100 WBC was binomially distributed with parameter  $\mu=2$ .
- We assumed that the number of bacteria colonies on 100 cm<sup>2</sup> agar plate was Poisson distributed with parameter  $\mu=2$ .
- We assumed that the distribution diastolic blood pressure measurement in 35-44 year old men was normal with mean  $\mu=80$  mm Hg.

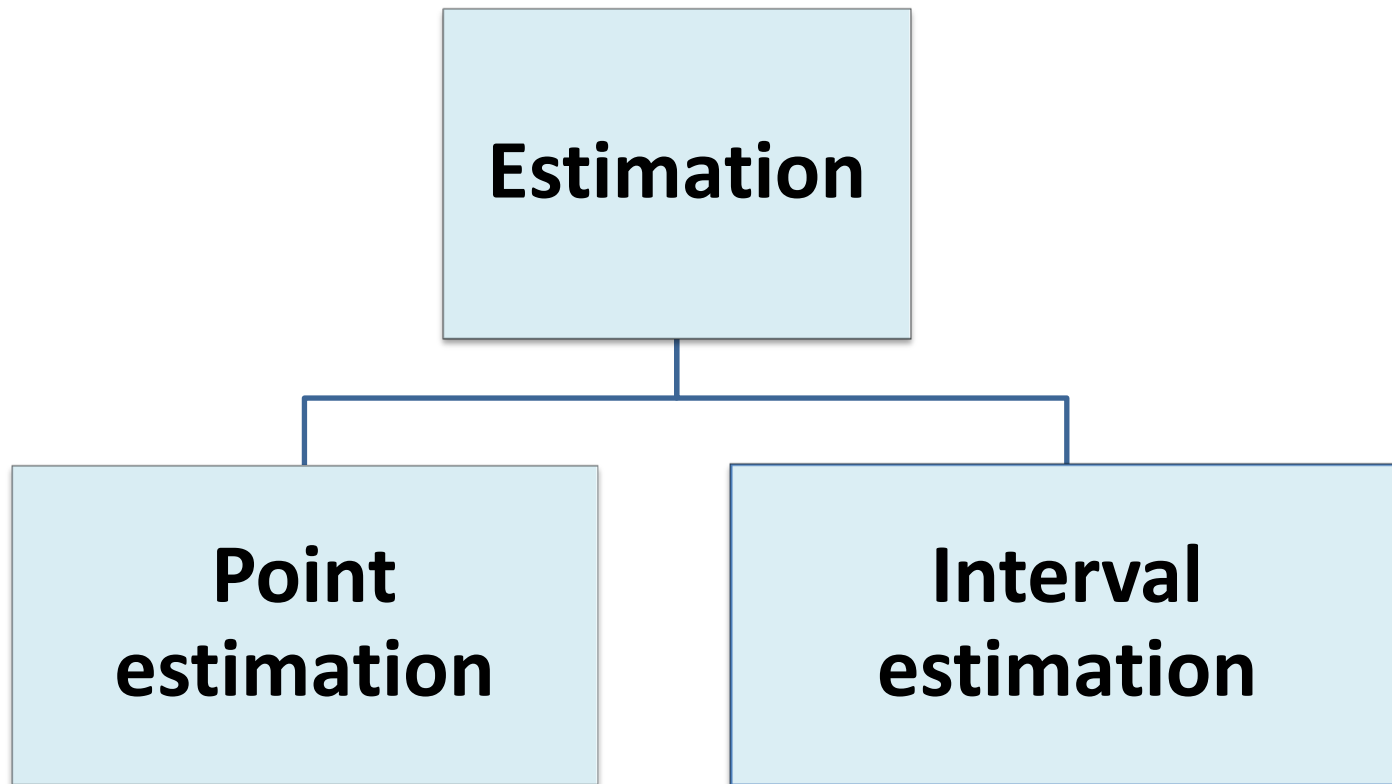
# Sample Statistics as Estimators of Population Parameters

A sample statistic is a numerical measure of a summary characteristic of a sample.

A population parameter is a numerical measure of a summary characteristic of a population.

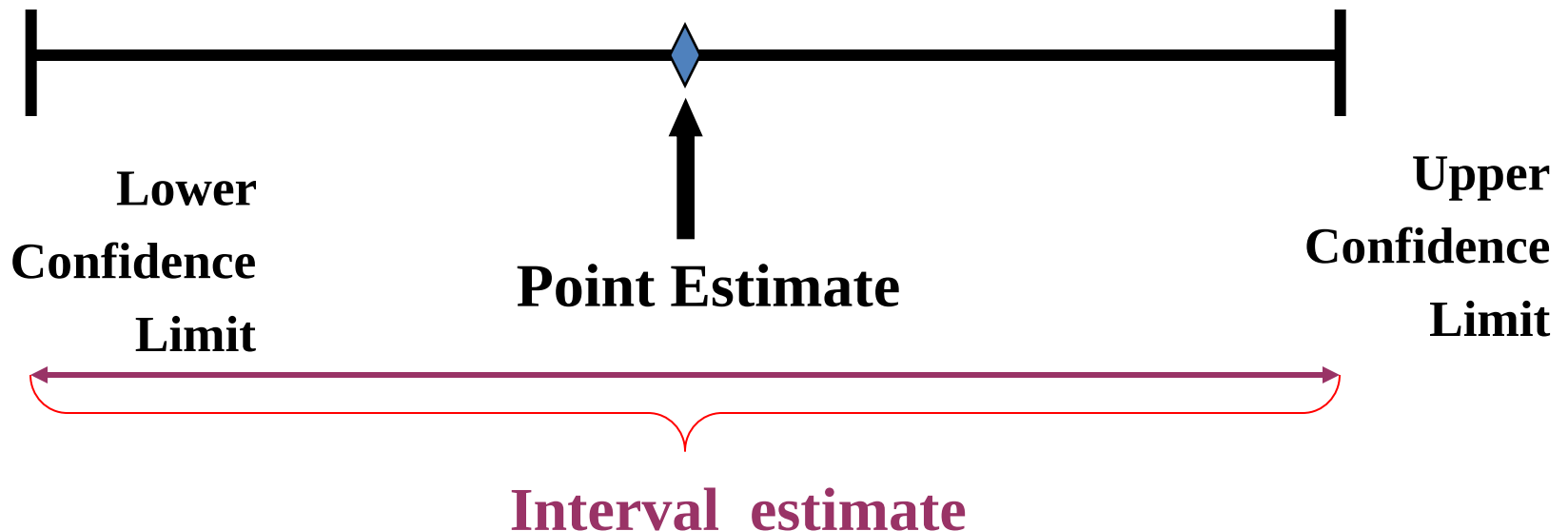
- An **estimator** of a population parameter is a sample statistic used to estimate or predict the population parameter
- An **estimate** of a parameter is a *particular* numerical value of a sample statistic obtained through sampling.

# Estimation



# Point and Interval Estimates

- A **point estimate** is a single value used as an estimate of a population parameter
- **Interval estimate** is a range or interval of numbers believed to include unknown population parameter with a certain degree of assurance
- *Point estimate is always within the interval estimate*



# Estimation Process

**Population**

Mean,  $\mu$ , is  
unknown

Random  
Sample

**Point estimate**

Mean

$\bar{X} = 50$

**Interval estimate**

I am 95%  
confident that  $\mu$   
is between 40 &  
60.



# Point estimation

- A single numerical value used to estimate the corresponding population parameter
- Gives little information about how close the value is to the unknown population parameter
- Example: Sample mean  $\bar{X}$  is **point estimate** of unknown population mean

# Sample statistic & their corresponding population parameter

**Statistic**

**Parameter**

|                     |           |           |            |
|---------------------|-----------|-----------|------------|
| Mean:               | $\bar{X}$ | estimates | $\mu$      |
| Variance:           | $s^2$     | estimate  | $\sigma^2$ |
| Standard deviation: | $s$       | estimate  | $\sigma$   |
| Proportion:         | $p$       | estimate  | $\pi$      |

**From sample**

**From entire population**

# Properties of good estimate

**a) Unbiasedness:** An estimator is said to be **unbiased** if its expected value is equal to the population parameter it estimates.

- For example: when  $E(\bar{X}) = \mu$ , the sample mean is an unbiased estimator of the population mean
- Unbiasedness is an average or long-run property. The mean of any single sample will probably not equal to the population mean, but the average of the means of repeated independent samples from a population will equal to the population mean.



# Properties of good estimate

## b) Minimum variance: (Efficiency)

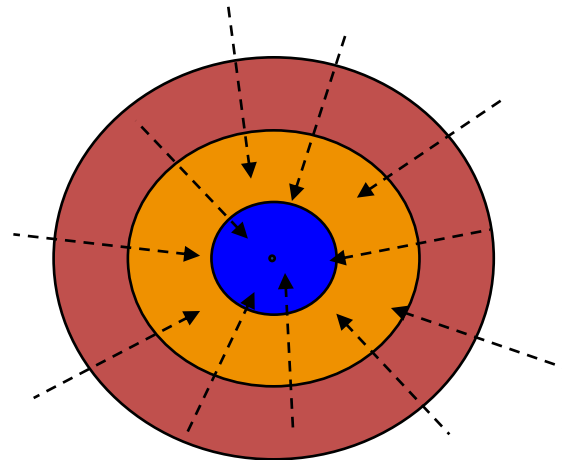
An estimate which has a minimum standard error is a good estimator

- For symmetrical distribution the mean has a minimum standard error and
- If the distribution is skewed the median has a minimum standard error

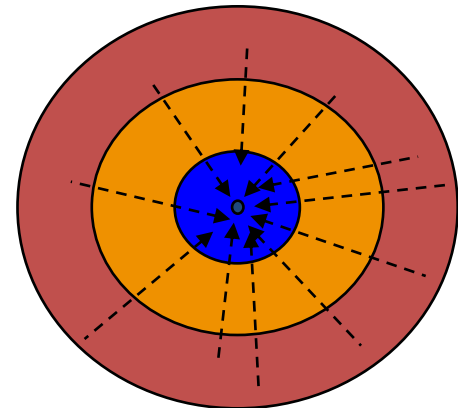
# Properties of good estimate

C) Consistency: An estimator is said to be **consistent** if its probability of being close to the parameter it estimates increases as the sample size increases

**Consistency**



$n = 10$



$n = 100$

# Interval estimation

- A single-valued estimate conveys little information about the actual value of the population parameter, about the accuracy of the estimate
- The probability of getting a sample statistic value that is exactly equal to the corresponding population parameter is usually quite small

# Interval estimation

- It is not reasonable to assume that a sample statistic value is exactly equal to the corresponding population parameter
- An interval estimate which locates the population parameter within an interval, with a level of confidence is needed

# Marginal Error

- A value calculated from the sample is the best guess when estimating corresponding population value
- Estimate is still uncertain due to sampling error
- Marginal Error is a measure of uncertainty
- Using marginal error you can state confidence interval: estimate  $\pm$  marginal error

# Confidence Interval or Interval Estimate

- **Confidence interval** or **interval estimate** is a range or interval of numbers believed to include an unknown population parameter
- Associated with the interval, there is a measure of the **confidence** we have that the interval does indeed contain the parameter of interest
- **Confidence interval**: provide a range of values of the estimate likely to include the “true” population parameter with a given probability

- A confidence interval or interval estimate has two components:
  - ✓ A range or interval of values
  - ✓ An associated **level of confidence**

# Confidence Level

1. Probability that the unknown population parameter falls within interval
2. Denoted  $(1 - \alpha)$  100%
  - $\alpha$  is probability that parameter is **not** within interval
3. Typical values are 99%, 95%, 90%

# Common Confidence Levels

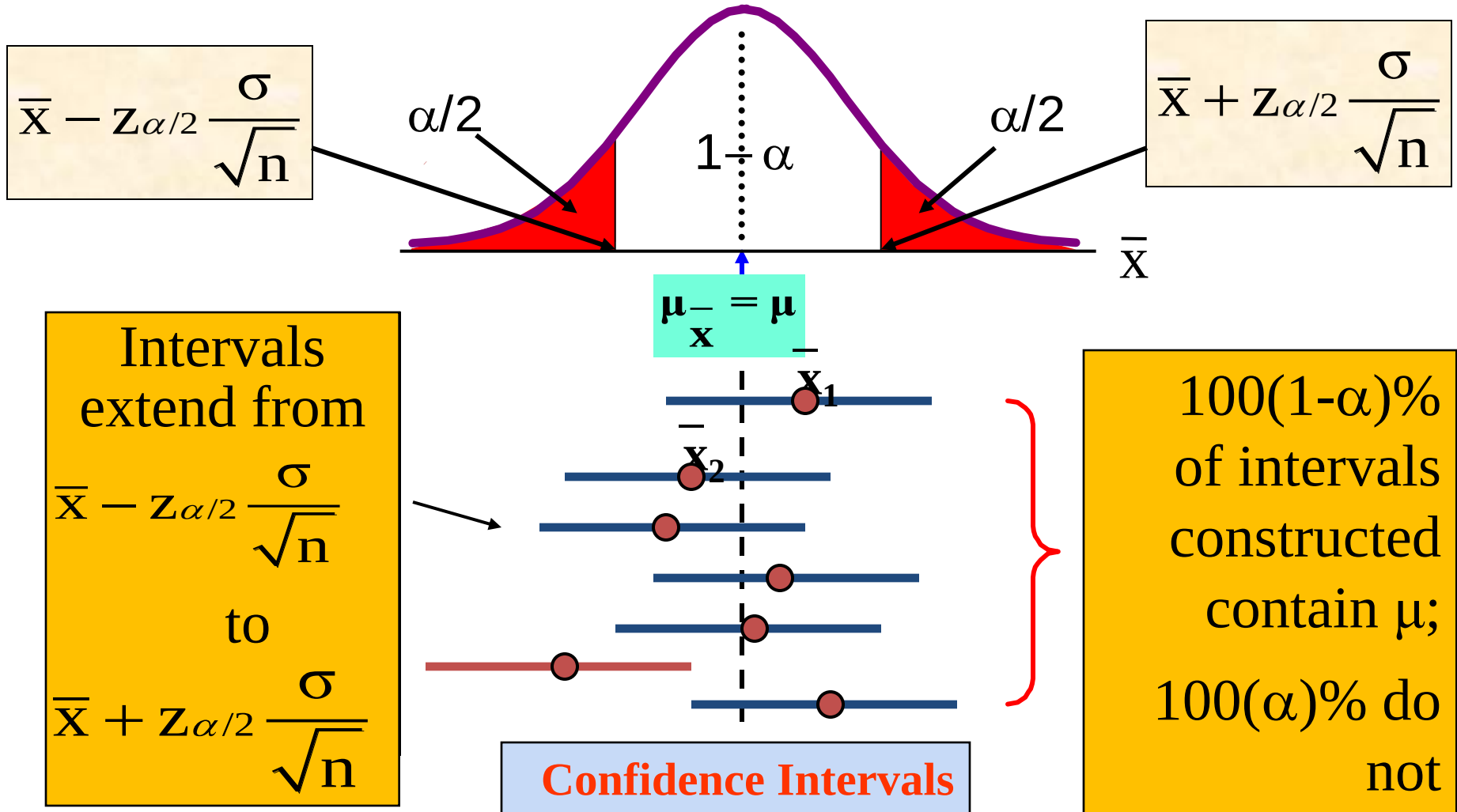
- Commonly used confidence levels are 90%, 95%, and 99%

| <i>Confidence Level</i> | <i>Confidence Coefficient, <math>1 - \alpha</math></i> | <i><math>Z_{\alpha/2}</math> value</i> |
|-------------------------|--------------------------------------------------------|----------------------------------------|
| 80%                     | .80                                                    | 1.28                                   |
| 90%                     | .90                                                    | 1.645                                  |
| 95%                     | .95                                                    | 1.96                                   |
| 98%                     | .98                                                    | 2.33                                   |
| 99%                     | .99                                                    | 2.58                                   |
| 99.8%                   | .998                                                   | 3.08                                   |
| 99.9%                   | .999                                                   | 3.27                                   |



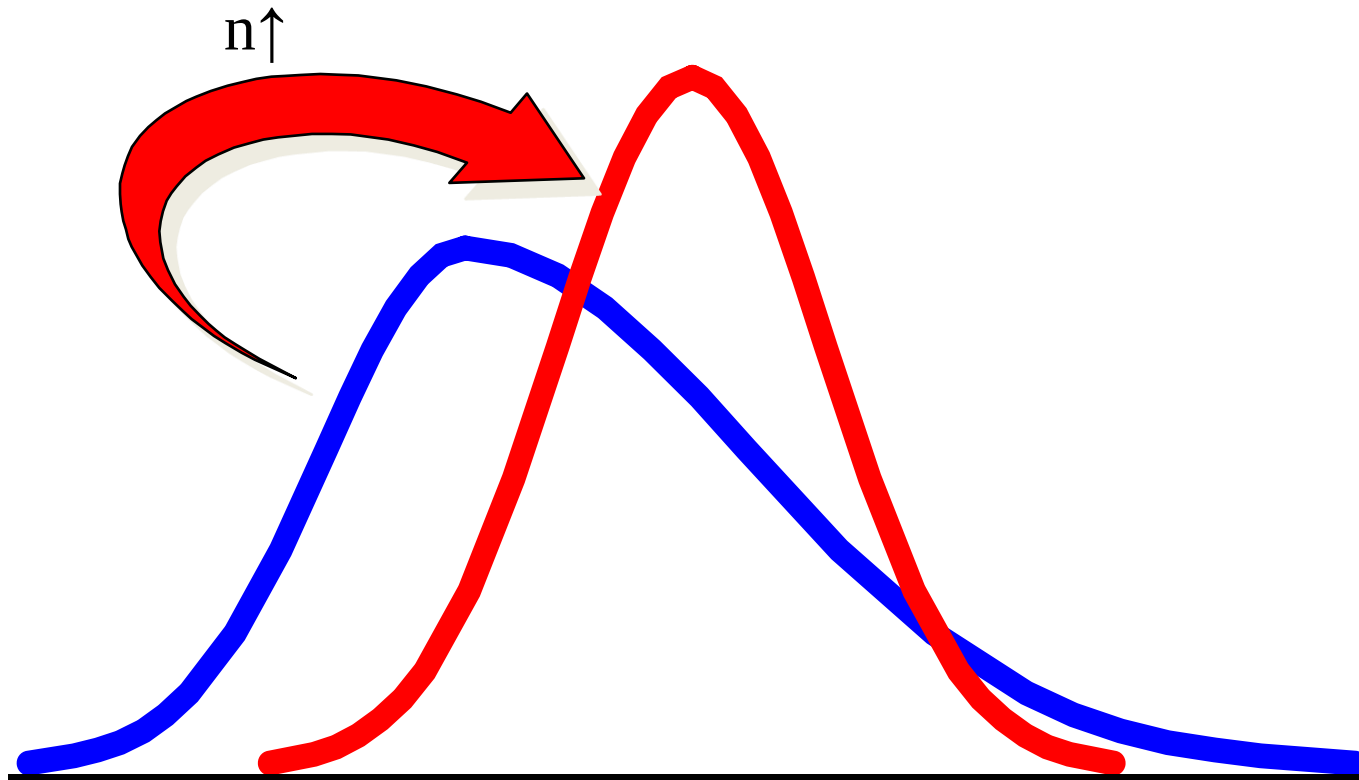
# Intervals and Level of Confidence

## Sampling Distribution of the Mean



# Central Limit Theorem

- As the sample size gets large enough, the sampling distribution becomes almost normal regardless of shape of population



# Central Limit Theorem

- **Rule of thumb:** Normal approximation for the sampling distribution will be good if  $n \geq 30$ . If  $n < 30$ , the approximation is only good if the population from which you are sampling is not too different from normal.
- Otherwise t-distribution

# CI for population mean: Large-sample size and when $\sigma$ is known

- For sufficiently large sample size  $n \geq 30$ , the sampling distribution of the sample mean, is approximately normal
- A  $100(1-\alpha)\%$  C.I. for  $\mu$  is:

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \Rightarrow \left( \bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

- $\alpha$  is to be chosen by the researcher, most common values of  $\alpha$  are **0.05**, 0.01, 0.001 and 0.1

# CI for population mean: Large-sample size and when $\sigma$ is unknown

- Whenever  $\sigma$  is not known (and the population is assumed normal), the correct distribution to use is the  $t$  distribution with  $n-1$  degrees of freedom. **However,** for large degrees of freedom, the  $t$  distribution is approximated well by the  $Z$  distribution
- A large sample  $100(1-\alpha)\%$  C.I. for  $\mu$  is:

$$\bar{X} \pm z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

- **Note that:** when  $\sigma$  is unknown,  $s$  is a good approximation of  $\sigma$

# Example

- An epidemiologist studied the blood glucose level of a random sample of 100 patients. The mean was 170, with a SD of 10. Construct the 95% CI for the population mean.

# Solution

$$\bar{X} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$170 \pm 1.96 \frac{10}{\sqrt{100}}$$

$$\Rightarrow (168.04, 171.96)$$

- We are 95% sure that the mean blood glucose level of the population lies between 168.04 and 171.96

## CI for population mean: small sample size ( $n < 30$ ) and when $\sigma$ is unknown

- If population standard deviation is unknown, then the sample means from samples of size  $n$  are t-distributed with  $n-1$  degrees of freedom
- A  $100(1-\alpha)\%$  C.I. for  $\mu$  is:

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$



# Example

- A study of hypoxemia during the immediate post-operative period reported the fractions of ideal weight for 11 patients who became severely hypoxemic during transfer to the recovery room. The mean is 1.51 and the standard deviation is 0.33. Estimate the 95% CI for the population mean fraction of ideal weight, where the population consists of hypoxemic patients similar to those in the study (use  $\alpha=0.05$ )

# Solution

$$t_{\alpha/2, n-1} = t_{0.025, 10} = 2.2281$$

$$1.51 \pm 2.2281 \frac{0.33}{\sqrt{11}}$$

$$1.51 \pm 0.221$$

$$(1.289, 1.731)$$

- We are 95% sure that the population mean fraction of ideal weight  $\mu$  lies between 1.289 and 1.731

# The t Table

| df  | t0.100 | t0.050 | t0.025 | t0.010 | t0.005 |
|-----|--------|--------|--------|--------|--------|
| 1   | 3.078  | 6.314  | 12.706 | 31.821 | 63.657 |
| 2   | 1.886  | 2.920  | 4.303  | 6.965  | 9.925  |
| 3   | 1.638  | 2.353  | 3.182  | 4.541  | 5.841  |
| 4   | 1.533  | 2.132  | 2.776  | 3.747  | 4.604  |
| 5   | 1.476  | 2.015  | 2.571  | 3.365  | 4.032  |
| 6   | 1.440  | 1.943  | 2.447  | 3.143  | 3.707  |
| 7   | 1.415  | 1.895  | 2.365  | 2.998  | 3.499  |
| 8   | 1.397  | 1.860  | 2.306  | 2.896  | 3.355  |
| 9   | 1.383  | 1.833  | 2.262  | 2.821  | 3.250  |
| 10  | 1.372  | 1.812  | 2.228  | 2.764  | 3.169  |
| 11  | 1.363  | 1.796  | 2.201  | 2.718  | 3.106  |
| 12  | 1.356  | 1.782  | 2.179  | 2.681  | 3.055  |
| 13  | 1.350  | 1.771  | 2.160  | 2.650  | 3.012  |
| 14  | 1.345  | 1.761  | 2.145  | 2.624  | 2.977  |
| 15  | 1.341  | 1.753  | 2.131  | 2.602  | 2.947  |
| 16  | 1.337  | 1.746  | 2.120  | 2.583  | 2.921  |
| 17  | 1.333  | 1.740  | 2.110  | 2.567  | 2.898  |
| 18  | 1.330  | 1.734  | 2.101  | 2.552  | 2.878  |
| 19  | 1.328  | 1.729  | 2.093  | 2.539  | 2.861  |
| 20  | 1.325  | 1.725  | 2.086  | 2.528  | 2.845  |
| 21  | 1.323  | 1.721  | 2.080  | 2.518  | 2.831  |
| 22  | 1.321  | 1.717  | 2.074  | 2.508  | 2.819  |
| 23  | 1.319  | 1.714  | 2.069  | 2.500  | 2.807  |
| 24  | 1.318  | 1.711  | 2.064  | 2.492  | 2.797  |
| 25  | 1.316  | 1.708  | 2.060  | 2.485  | 2.787  |
| 26  | 1.315  | 1.706  | 2.056  | 2.479  | 2.779  |
| 27  | 1.314  | 1.703  | 2.052  | 2.473  | 2.771  |
| 28  | 1.313  | 1.701  | 2.048  | 2.467  | 2.763  |
| 29  | 1.311  | 1.699  | 2.045  | 2.462  | 2.756  |
| 30  | 1.310  | 1.697  | 2.042  | 2.457  | 2.750  |
| 40  | 1.303  | 1.684  | 2.021  | 2.423  | 2.704  |
| 60  | 1.296  | 1.671  | 2.000  | 2.390  | 2.660  |
| 120 | 1.289  | 1.658  | 1.980  | 2.358  | 2.617  |
| ∞   | 1.282  | 1.645  | 1.960  | 2.326  | 2.576  |

# CI for a population proportion: Large-sample size

- For sufficiently large samples, the sampling distribution of the proportion  $p$  is approximately normal
- A  $100(1-\alpha)\%$  CI for  $\pi$  is:

$$p \pm Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

A sample is considered large enough when both  $n \cdot p$  and  $n \cdot q$  are greater than 5, where  $q = 1 - p$ .

# Example

- A study on dental health practice. Of 300 adults interviewed, 123 said that, they regularly had a dental check-up twice a year. What is the 95% C.I. for  $\pi$ ?

$P = 123/300 = 0.41$  a point estimator of  $\pi$

$$\alpha = 0.05 \Rightarrow Z_{0.025} = 1.96$$

$$p \pm Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

$$0.41 \pm 1.96 \sqrt{\frac{(0.41)(0.59)}{300}}$$

$$\Rightarrow (0.3543, 0.4656)$$

**Interpretation: ?**

**Thank you**